

I've Got a Feeling: Sentiment analysis for collaborative performance in VR

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ABSTRACT

Sentiment analysis is a common method for evaluating the meaning of text. While commonplace, the application of sentiment analysis for assessing collaborative teamwork is limited. We present an exploratory study of sentiment analysis as a useful companion to established metrics of user collaboration in VR. We present results of paired users ($N=14$, $N_{pairs}=7$) completing a VR brainstorming task, and sentiment analysis on their conversation. We report on how the sentiment process maps to the Team Workload Questionnaire (TWLQ) and illustrate how this could be a valuable metric for supporting an enriched evaluation of collaborative VR.

Index Terms: VR, Sentiment analysis, conversational analysis, team working, collaboration

1 INTRODUCTION

Sentiment analysis is the process of extracting the meaning within a body of text and is commonly applied within systems to determine a measure of positivity or negativity. While, sentiment analysis is frequently applied to human authored text (i.e., online product reviews, questionnaire responses) [6] and therefore prolific in numerous fields (notably in e-commerce [7]), the use of sentiment analysis within immersive systems is still limited. Slater et al. [5] argue for the use of sentiment analysis in XR for evaluations. Additionally, they highlight current evaluation methods could be combined with sentiment analysis for a deeper insights and understanding into user behaviour and experience. While some work has also explored the extraction of human physiological measures in VR and AR [1] in single user interfaces, limited work has considered how the sentiment in a user conversation could be applied for assessing the quality of an collaborative experience. To begin addressing this, we present an initial study on the application of sentiment analysis in a collaborative VR task. We present the classical method for sentiment analysis based on dictionaries of positive, negative, and stop words, and how this approach can be applied in the analysis of user conversation within collaborative VR. We also explore the initial adaptation of the dictionaries in the sentiment process, and highlight optimisation and their link to common metrics like TWLQ.

2 RELATED WORK

2.1 Sentiment Analysis

Sentiment is a personal feeling or opinion and is based on human emotion. Complementary to this, the process of extracting personal opinion for analysis is the fundamental aim of sentiment analysis [6]. Sentiment analysis is commonplace in many domains of text analysis and has applications in e-commerce evaluation [7], user response analysis [2] and text mining [3].

2.1.1 Positivity and Negativity

Sentiment analysis is commonly defined as a measure of the total number of positive and negative words in a passage of text expressed as the *Polarity Score* (PS) and the *Subjectivity Score* (SS).

PS and SS are defined in equations 1 and 2, where $|PW|$ is the total positive words, $|NW|$ is the total negative words, $|TW|$ is the total overall words and λ is a constant set to 0.000001. PS occupies a scale range between -1 to $+1$, where a -1 indicates complete negativity, a $+1$ indicates absolute positivity and 0 indicates neutrality. SS occupies a range from 0 to 1 indicating subjectivity level.

$$PS = \frac{|PW| - |NW|}{(|PW| + |NW|)} + \lambda \quad (1) \quad SS = \frac{(|PW| + |NW|)}{|TW|} + \lambda \quad (2)$$

2.1.2 Word Dictionaries

When calculating PS and SS word dictionaries are used. These dictionaries contain a corpus of both positive words (i.e., “wonderful”, “glad”, “happy”), negative words (i.e., “wrong”, “sick”, “impede”, “broken”) and stop words, which represent neutral or pauses (i.e., “am”, “an”, “I”, “like”, “yes”). While these dictionaries are commonly designed for general applications of sentiment, they also contain words common to user performance of systems (i.e., “buggy”, “lag”, “judder”, “enjoy”, “viewable”, “seamless”) (see Section 3.2 for more detail). The classical process as mentioned first extracts the stop words from the passage of text using the stop word dictionaries before calculating PS and SS meaning that any words within the stop word dictionaries are not considered in the sentiment calculation.

3 METHODOLOGY

We conducted a pilot study to explore sentiment analysis in collaborative VR, where 14 Users (8 male, 5 female, mean age 25) were paired into partnerships ($N=7$). The participants were tasked with a brainstorming task.

3.1 Environment, Equipment and Task

For the study, users were co-located, each had their own space and were separated from their partner via a dividing screen (see Fig. 1). Users were tasked to complete brainstorming using the Noda App¹ on a Meta Quest 2 device. Noda supports real-time collaboration, thus both participants wore a Quest 2 VR device (see Fig. 1) and created “nodes” (cubes or spheres) and labelled according to the task. Two topics were chosen for the brainstorming task of comparable complexity, namely “the advantages and disadvantages of mobile phones” and “the advantages and disadvantages of remote working”. Participant pairs had 10 minutes to complete the task with topics randomized across pairs.

3.2 Process

Users verbalised their thoughts and were recorded via the Meta Quest. Using common datasets of positive, negative and stop words² we report on sentiment analysis *without stop word removal* (P1: PS1, SS1, TW1) and *with classical stop word removal* (P2: PS2, SS2, TW2). We also report on the removal of *non-VR experience stop words* only (P3: PS3, SS3, TW3). To achieve this, we

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¹<https://noda.io/>

²<https://www.kaggle.com/datasets/vivek603/text-analysis?resource=download>

Pair	PS1	SS1	TW1	PS2	SS2	TW2	PS3	SS3	TW3	TWLQ1	TWLQ2	$\bar{x}$
1	0.18	0.03	1941	0.11	0.09	529	0.15	0.08	575	51.88	38.75	45.31
2	0.16	0.03	1707	0.08	0.08	462	0.13	0.08	520	34.38	40.63	37.50
3	0.52	0.02	1220	0.43	0.06	346	0.45	0.06	396	55.00	46.88	50.94
4	0.16	0.02	1575	0.03	0.07	425	0.09	0.07	487	37.50	64.38	50.94
5	0.33	0.02	2138	0.24	0.06	582	0.28	0.06	656	38.13	48.75	43.44
6	-0.02	0.03	1361	-0.20	0.10	348	-0.11	0.09	423	43.13	36.25	39.69
7	-0.04	0.03	3017	-0.18	0.08	823	-0.13	0.08	912	41.25	45.65	43.44

Table 1: Results for the three sentiment analysis processes (P1, P2, P3). The polarity score, subjectivity score, and total words are given *without stop word removal* (PS1, SS1, TW1), *with classical stop word removal* (PS2, SS2, TW2) and with the removal of *non VR experience stop words only* (PS3, SS3, TW3). Team workload is given for each individual user in the pair (1 and 2) alongside the $\bar{x}$ for TWLQ.

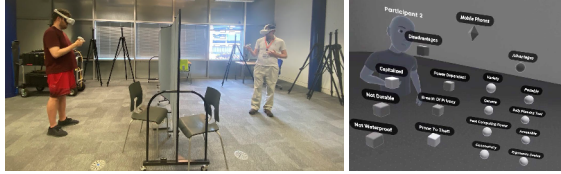


Figure 1: The environment for the collaborative brainstorming task in VR (left) and the interface for the Noda application (right).

created a new stop word dictionary tailored specifically to VR systems. Often stop word dictionaries may contain valuable data useful for XR system analysis, and removing these could reduce VR sentiment analysis reliability. For this we compiled a new stop word dictionary where stop words which are not relevant to the VR experience (i.e., “no”, “me”, “I”, “like”, “yes”) were removed and words relating to *task* (i.e., “up”, “down”, “left”, “right”), *user experience* (i.e., “anybody”, “me”, “self”), *interaction* (i.e., “look”, “see”, “engagement”) and *interfaces* (i.e., “enter”, “next”, “input”) were not removed. We also captured team workload via the TWLQ [4] providing further insights in user collaboration and task performance.

4 RESULTS

Table 1 illustrates the results for the three sentiment analysis processes (P1, P2 and P3). We perform Pearson’s linear correlation with an ($\alpha = 0.05$) for the team-level sentiment (PS and SS) and average team workload $\bar{x}$ TWLQ (see Figure 2). Regarding PS, a significant positive correlation was found with $\bar{x}$ TWLQ ($r = 0.49, p < 0.02$), however, for the sentiment processes (P1, P2 and P3) no significant effects were reported against ($\bar{x}$) TWLQ for P1: PS1 ($r = 0.54, p = 0.21$), P2: PS2 ($r = 0.49, p = 0.26$) and P3: PS3 ($r = 0.49, p = 0.26$). Regarding SS a significant negative correlation was found between the sentiment processes $\bar{x}$ TWLQ ($r = -0.49, p < 0.03$) with moderate strength. Where, SS revealed a significant negative correlation between P2: SS2 ($r = -0.57, p = 0.04$) and P3: SS3 ($r = -0.63, p = 0.02$) and $\bar{x}$ TWLQ with a strong effect. No significant effect was reported for P1: SS1 ($r = -0.72, p = 0.06$).

5 DISCUSSION AND CONCLUSION

The initial results presented in Table 1, Figure 2 and Section 4 highlight a positive correlation between the overall polarity score (PS) and the mean Team Workload Questionnaire ($\bar{x}$ TWLQ) with a moderate strength. However, no significant effect was reported for the individual sentiment analysis processes (P1, P2 and P3). Additionally, a moderate negative correlation found in the subjectivity score (SS) against the $\bar{x}$ TWLQ for the sentiment processes. For processes P2 and P3 there was a stronger inverse correlation, in contrast to P1 with no significant effect. Initial results supports the proposition of user utterances during the tasks can reflect team workload and can influence team working, where higher subjectivity scores can relate

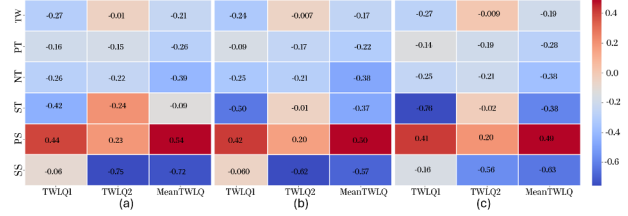


Figure 2: Pearson correlation analysis of each sentiment process *without stop word removal* P1 (a) *with classical stop word removal* P2 (b) and *with non VR experience stop word removal* P3 (c).

to more bias in conversations and thus decreasing collaboration in VR. Although comparable relationships may exist in the different implementations of sentiment analysis for both PS and SS, the removal of stop words demonstrated a stronger negative correlation in SS against $\bar{x}$ TWLQ in the P3 process (the extraction of *non VR experience stop words only*). These preliminary results highlight the value of sentiment analysis in measuring user and team experiences in collaborative VR. Future work includes refining VR-specific dictionaries to better capture user experiences, exploring the application of sentiment analysis to manual tasks, investigating collaboration in larger team settings, and incorporating speaker diarisation to analyse patterns in sentiment of collaborating users.

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